

data over a network and so on. With the rise in digital music players and streaming services, there has been a demand among the audiophiles and enthusiasts for better compression methods, lossless audio and more clarity in the audio files.

Audio compression and the dawn of the 1s and 0s.

Digital files are full of 1s and 0s. When a file is written, the method of writing is not linear, which means that the data is not physically continuous, but logically continuous. This means that if you read the file in an order you were not supposed to, all you would find is garbage. The problem with this is that, there are a lot of redundant spaces in between the data written. These gaps need to be represented in the file to maintain the sequential order and hence cannot be randomly eliminated. Compression solves this by using specific algorithms to remove the spaces and group all the data together. The algorithms also find patterns in the data which are repeated, so that they can be represented in a manner which consumes less data.

While compression is good to save space and bandwidth, the algorithms are not completely accurate. The method in which the data is recreated is not perfect, which will lead to the dropping of some blocks of data. Since the devices process the compressed files and read the data, they wouldn't be able to restore the lost data and this causes noisy and inconsistent audio files.

Compressed and non compressed audio files have different file formats, MP3 is the most common compressed audio format and FLAC is the best uncompressed format. The goal of most audio formats is to recreate the original audio stream exactly, without any losses after various conversions by the DAC. The quality of the decompressed music is measured in bitrates.

What is the bit rate?

Bitrates represent the amount of data which is stored per unit time in a file format. An MP3 file with a high bitrate stores more audio data per unit second than ones with lower bitrates. Bitrates determine the quality of an audio file. The industry standard bitrate for MP3 files is considered to be 320 kbps. This is a let-down compared to other formats of music storage such as CDs, which support bit rates as high as 1411 kbps. This simply means that there is more information about the sound being produced by the file, which allows more details to be created when reproducing the sound. The audio from higher bitrate files sound clearer, and some even let you distinguish the individual audio sources such as voices and musical instruments.